Automatic RFI Identification for Sentinel-1 Based on Siamese-Type Deep CNN Using Repeat-Pass Images

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***Abstract*— Since the start of the Sentinel-1 (S-1) mission, numerous cases of severe image degradation caused by radio frequency interference (RFI) have been reported, which puts forward an urgent need for RFI identification and mitigation. In this article, an automatic RFI identification method is proposed based on a Siamese-type deep convolutional neural network (Siam-CNN-RIM). The Siam-CNN-RIM can be served as a preprocessing step before RFI mitigation to identify whether an S-1 image is RFI-contaminated or not. Different from traditional RFI identification networks which only use a single image as input, an additional image in the repeat-pass time series is also fed into the input of Siam-CNN-RIM as a reference. Both the input images correspond to the same illuminated area and pass through the same convolutional layer followed by an energy function such that the different features caused by RFI can be extracted and the background terrain features can be ignored. This is beneficial for distinguishing the real RFI signatures and the similar terrain signatures that may cause false positives and thus improving the RFI identification performance. Experimental results show that the proposed method is robust in different scenarios and can achieve more than 97% RFI identification accuracy, even for the open-set task where the test scenarios are not included in the training set.**

***Index Terms*— Automatic radio frequency interference (RFI) identification, deep convolutional neural network (CNN), repeat-pass images, Sentinel-1 (S-1), Siamese network.**

I. Introduction

A. Background

SYNTHETIC aperture radar (SAR) is an active remote sensing technology in both civil and military areas since it can operate day and night in nearly all weather conditions [1]. With the rapid growing development of radio technology, the finite electromagnetic environment is becoming increasingly crowded. The received signals of the SAR systems are interfered by some unwanted signals [generally referred to as radio frequency interference (RFI)], which are radiated by other equipment sharing the same operating frequency band, such as noncooperative radar systems, AM/FM radio, global navigation satellite systems (GNSS), and other communication systems [2], [3]. The RFI has caused severe contamination to the images of various SAR systems operating at different wave bands, including P-band EcoSAR, L-band ALOS-1 PALSAR, and X-band TerraSAR [4]–[6].

The contamination of RFI is also a severe problem in C-band Sentinel-1 (S-1) SAR launched by the European Space Agency (ESA) [7]–[10]. The image degradation caused by RFI has aroused the concerns of the users since the start of the S-1 mission. The RFI causes haze-like artifacts or bright lines in the S-1 image and has several side effects on the image interpretation process. For example, ESA has observed the image degradation caused by the mutual interference between S-1A/B, Radarsat-2, and Gaofen-3 [6], [8], [11]. The contamination from ground-based radars has also been demonstrated in [12]. As reported in [9] and [13], the signature of RFI poses a hindrance for ship detection.

Since the SAR system is severely affected by the RFI, interference suppression is a crucial task and has been a hot topic in the past decades. Fruitful results have been achieved in the RFI suppression field, by using filtering [14]–[17], blind source separation [18], [19], and low-rank/sparse recovery techniques [20]–[35]. In addition to RFI suppression, the automatic RFI identification is also an important task [18], [19], [36]. On the one hand, the RFI identification is an indispensable preprocessing step before RFI suppression. Without RFI identification, the RFI-free images would also be processed by the RFI suppression method, which leads to the degradation of the image quality and the waste of valuable computing resources. On the other hand, since the SAR satellite continuously monitors the Earth's surface day and night with a short revisit period, a bulk of data is produced. As shown in the database provided by the Alaska Satellite Facility, the S-1 produces up to 4 081 577 Level-1 images from 2014 to September 14, 2021 [37]. Manual evaluation of this huge amount of images is time-consuming and even boring. These arouse an urgent need for an algorithm that can perform the RFI identification efficiently and automatically.

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B. Related Work

During the past decade, various RFI identification methods have been proposed, and they can be divided into three categories.

The first category is based on the echoes of the illuminated area in the raw data domain. Zhou *et al.* [18] proposed an RFI identification method with a threshold in the frequency domain of raw data. Assuming that the RFI would change the statistical distribution of the echoes, Tao *et al.* [19] used statistical quantities (e.g., Kurtosis) to discriminate RFI-contaminated images. An RFI identification method in the time–frequency domain was proposed in [38]. These methods have good performance when the power of RFI is much higher than the echoes of illuminated scene but suffer performance degradation when the RFI power is weak or comparable to the echoes.

The second category is based on sniffing pulses. In [10], for the S-1 data in interferometric wide-swath (IW) mode, the first 8–10 received pulses of each burst, which are unaffected by the background scattering, are used for spectrum monitoring and RFI identification. A time-bandpass filter was proposed in [4] on the sniffing pulse interleaved during the EcoSAR operation. These methods sniff the electromagnetic environment at the absence of background echoes and therefore possess high sensitivity and can achieve good performance even when the RFI is weak. However, since the RFI identification is only performed during the interval of the received echoes of SAR, it is likely that the transiently occurred pulsed RFI fails to be detected. Moreover, the first two categories are performed in the raw data domain, while the level-0 SAR raw data are often unavailable to users [12]. These facts give the rise to the demand for the third category of RFI identification methods.

The third category is performed directly on Level-1 images. Based on the S-1 ground range detected (GRD) products, an RFI identification method is proposed in [36] by using the polarization characteristics. This method is effective in most situations where the RFI is much stronger in the cross polarization than in the co-polarization channel but would cause leak detection if such an assumption is not valid. Based on S-1 single-look complex (SLC) quick-look images, a manual feature extraction method was proposed in [39], with the use of pixel convolution, thresholding, and nearest neighbor structure filtering. Although this method shows satisfying performance in some particular situations, the feature extraction algorithm has to be manually designed for different scenes. To handle this problem, an automatic RFI identification method was proposed in [40] by using a LeNet-type convolutional neural network (CNN). The feature extraction and RFI identification process are performed automatically by the convolutional layer and the classifier, respectively. However, the generalization ability of the model is not sufficient, and the network still has to be individually designed and trained for different scenes, which poses a hindrance for the application of this method in practice.

C. Our Work and Contributions

In this article, we aim to provide a feasible general solution to the automatic RFI identification problem for S-1, which is robust to different scenes. We deal with such problem based on the IW mode GRD products of S-1. Level-1 GRD products consist of the focused SAR data that have been detected, multilooked, and projected to the ground range using an Earth ellipsoid model. Unlike the Level-0 raw data products, the Level-1 products are operationally available and free of charge to all users [41]. Moreover, more features of terrain surface and RFI signature can be observed in Level-1 data than in Level-0 data, especially when the RFI power is not significantly larger than the echoes. The Level-1 products of S-1 contain SLC products and GRD products. Compared with the SLC product, the GRD product is in the ground range coordinate system and has an advantage of reduced speckle noise due to multilooked processing. IW mode data are used here among the various acquisition modes of S-1 since IW mode has a wider coverage over land than other acquisition modes of S-1 [40]–[42], which means that a larger dataset can be achieved. Moreover, in the IW mode, various repeat-pass images corresponding to the same illuminated region are available. It is beneficial for improving the RFI identification performance, which we would analyze in detail in the remainder of this article.

As mentioned above, the usage of GRD data has a lot of advantages over other S-1 data products, but it would also bring about difficulties. The diversities of terrain types and RFI signatures in the GRD image pose a great challenge for the establishment of a generalized RFI identification model. One typical problem is that the features of some terrain scatterers are quite similar to the radiometric artifacts of RFI in GRD images, which may easily lead to the false alarm of the RFI detector. Considering that the S-1A/B has a fixed revisit duration with 12-day repeat cycle for an orbit, where different repeat-pass acquisitions show similar features of terrain surface but time-varying RFI signatures, the correlation and coupling information among different repeat-pass GRD images are utilized in this article by a well-designed Siamese-type deep CNN for RFI identification (Siam-CNN-RIM). Different from traditional CNN-based RFI identification networks which only use a single image as input, an additional image in the repeat-pass time series is also fed into the input of Siam-CNN-RIM. These two image inputs correspond to the same illuminated area and are passed through the same convolutional feature extraction layer. Afterward, the extracted features of both input images are compared and the decision is made by an energy function. As such, the proposed method can focus on detecting and extracting the features of RFI with a reduced influence of the terrain surface diversities, thus avoiding the overfitting problem and improving the generalization ability in different illuminate scenes. The main contributions of this article can be summarized as follows.

1) The general analytical form of the RFI in SAR images is derived, which is fundamental for the theoretical analysis of the common characteristics of different RFIs.
2) The survey of current RFI situation in S-1 repeat-pass GRD data is provided. The shape, intensity, and polarization characteristics of RFI are analyzed. The similarities and differences of the terrain surface and

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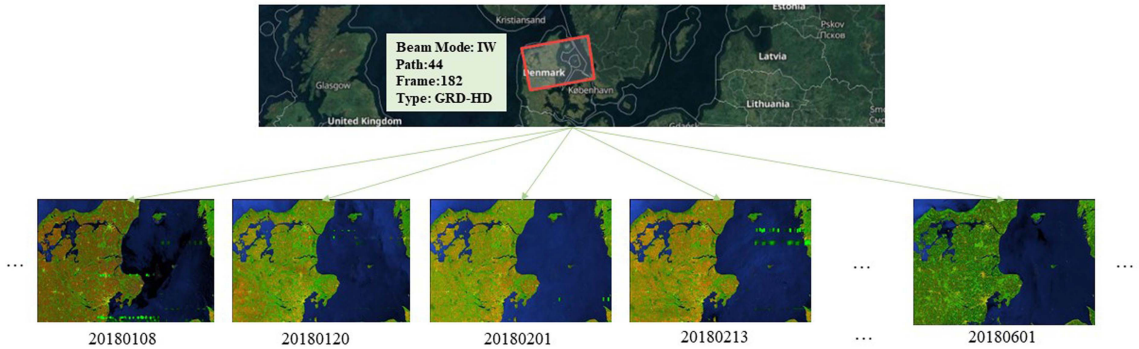


Fig. 1. Example of the repeat-pass GRD images of S-1 (orbit index: Frame 182 and Path 44).

the RFI signature in the repeat-pass GRD images are investigated.

3) An automatic RFI identification method based on S-1 repeat-pass GRD images, which is robust to different scenes, is proposed. A tailor-made Siamese-type deep CNN is designed to enhance the RFI identification performance. As far as we know, this is the first published work that utilizes the repeat-pass GRD images for the RFI identification of S-1.
4) A well-designed data augmentation technique is proposed for the training of the Siam-CNN-RIM model. Data augmentation to avoid overfitting allows us to implement a deeper network than state-of-the-art RFI identification methods. In addition to the imposition of the Siamese structure, the data augmentation and a deeper CNN are a plus to enhance the RFI identification performance.
5) Extensive experiments are conducted to evaluate the proposed method in various scenes with different terrain types and RFI signatures. The results show that the proposed method outperforms existing methods, in which the model trained by a few typical scenarios is robust for the RFI identification in other scenarios.

The remainder of this article is organized as follows. Section II presents the characteristics of terrain surface background and RFI in S-1 repeat-pass GRD images. Then, the automatic RFI identification method is proposed in Section III. Experimental results are shown in Section IV. Section V makes a discussion on the proposed method. Finally, this article concludes with a brief summary in Section VI.

II. Characteristics of RFI in S-1 Repeat-Pass GRD Images

As we know, each of both S-1 satellites (i.e., S-1A and S-1B) is in a near-polar, sun-synchronous orbit, with a 12-day repeat cycle and 175 orbits per cycle, which ensures that a single S-1 satellite is able to revisit the same region of landmass in the IW mode once every 12 days. Fig. 1 shows an example of the S-1A repeat-pass GRD high-resolution dual-polarization images (index of orbit: Frame 182 and Path 44). This frame mainly covers the area of Denmark and part of the images is shown in the second row of Fig. 1. These images have similar terrain surface background, but varying RFI signatures. In this section, the analytical expressions of the RFI patterns in SAR images are derived. The characteristics of RFI in a single GRD image are analyzed, including its shape, intensity, and polarization. Then, the additional information that multiple repeat-pass GRD images can provide is analyzed.

A. Generation Mechanism of RFI in S-1 Images

For the raw complex-valued echo data $X(t,\eta)$, where t and η denote the fast time and slow time, respectively, it is composed of three components: the SOI $S(t,\eta)$, the RFI $I(t,\eta)$, and the noise $N(t,\eta)$

$$X(t,\eta) = S(t,\eta) + I(t,\eta) + N(t,\eta). \tag{1}$$

The transmitted signal of S-1 can be written as

$$S_{tr}(t) = \text{rect}\left(\frac{t}{T}\right)\exp\left(j\pi K t^2\right) \tag{2}$$

where K is the chirp rate. The RFI can be represented as a general form with varying amplitude $a(t)$ and phase $\mu(t)$

$$I_{tr}(t) = a(t)\exp[j\mu(t)] \tag{3}$$

where $\mu(t)$ is a general time-modulated term. It can be a mono-frequency narrowband component or a frequency-modulated wideband term. Then, the RFI received by the SAR system can be written as $I(t,\eta)$ that can be denoted as

$$I(t,\eta) = \gamma\,\delta(\eta)a(t-\tau_0)\exp[j\mu(t-\tau_0)] \tag{4}$$

where γ denotes the intensity of RFI; $\delta(\eta)$ is a Dirac function, which indicates that the RFI is received when $\eta = 0$; and τ_0 denotes the RFI delay with respect to the reference time of a pulse observation interval.

To model the RFI patterns in the image, the closed-form expression of the RFI response is derived by adopting typical SAR imaging algorithms. Without loss of generality, the omega-K algorithm is illustrated as an example to derive the general form of the image-domain RFI.

In the omega-K algorithm, the received signal of SAR has to be transformed into the wavenumber domain via 2-D Fourier transforms. However, for the RFI with more than two-order fast-time-varying phase term, no closed form of the range-frequency spectrum can be derived. Therefore, in the following, we would mainly focus on the derivation of the slow-time analytical expressions of the image-domain RFI.

To perform omega-K algorithm, the first step is to transform the received RFI signal of (4) into wavenumber domain via

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2-D Fourier transforms. The RFI in wavenumber domain can be written as

$$I_1(f_t, f_\eta) = \gamma_1 p_1(f_t) \tag{5}$$

where f_t and f_η denote the range and azimuth frequencies, respectively, γ_1 denotes the intensity, and $p_1(f_t)$ denotes the range-frequency component.

The second step of omega-K algorithm is to multiply the wavenumber-domain spectrum by a reference function $Q(f_t, f_\eta)$

$$Q(f_t, f_\eta) = \text{rect}\left(\frac{f_\eta - f_{\eta c}}{f_{\text{PRF}}}\right)\exp(j\theta_{\text{ref}}) \tag{6}$$

where f_{PRF} denotes the processed pulse repetition frequency (PRF), $f_{\eta c}$ denotes the central Doppler frequency, and $\exp(j\theta_{\text{ref}})$ denotes

$$\theta_{\text{ref}} = \frac{4\pi R_{\text{ref}}}{c}\left[\frac{f_t}{D(f_{\eta c}, V)} + f_0 D(f_{\eta c}, V)\right] + \frac{\pi f_t^2}{K_r} \tag{7}$$

where V denotes the effective velocity of the SAR platform, c denotes the speed of light, and f_0 denotes the carrier frequency. $D(f_{\eta c}, V)$ is a hyperbolic range cell mitigation parameter defined as

$$D(f_{\eta c}, V) = \sqrt{1 - \frac{c^2 f_\eta^2}{4V^2 f_0^2}}. \tag{8}$$

After the second step of the omega-K algorithm, the RFI can be written as

$$I_2(f_t, f_\eta) = \gamma_2 \text{rect}\left(\frac{f_\eta - f_{\eta c}}{f_{\text{PRF}}}\right)\exp\left[j\theta_2(f_\eta)\right] \cdot p_2(f_t) \tag{9}$$

where f_t and f_η are uncoupled, γ_2 denotes the intensity, and $p_2(f_t)$ denotes the range-frequency term. $\theta_2(f_\eta)$ is

$$\theta_2(f_\eta) = \frac{4\pi R_{\text{ref}} f_0 D(f_{\eta c}, V)}{c}. \tag{10}$$

In the third step of omega-K algorithm, $I_2(f_t, f_\eta)$ is transformed into range-Doppler domain by performing inverse fast-time Fourier transform and then is multiplied by the azimuth matched filtering function $\exp(j\theta_{\eta m})$, where $\theta_{\eta m} = 4\pi(R_0 - R_{\text{ref}}) f_0 D(f_{\eta c}, V)/c$. The RFI after the third step can be written as

$$I_3(t, f_\eta) = \gamma_3 \text{rect}\left(\frac{f_\eta - f_{\eta c}}{f_{\text{PRF}}}\right)\exp\left[j\theta_3(f_\eta)\right] \cdot p_3(t) \tag{11}$$

where γ_3 denotes the intensity, $p_3(t)$ denotes the fast-time term, and $\theta_3(f_\eta)$ is

$$\theta_3(f_\eta) = \frac{4\pi R_0 f_0 D(f_\eta, V)}{c}. \tag{12}$$

Then, in the fourth step, $I_3(t, \eta)$ is processed by the inverse slow-time Fourier transform, and the image-domain RFI $I_4(t, \eta)$ is finally obtained. With the use of the principle of stationary phase (POSP), $I_4(t, \eta)$ can be written as

$$I_4(t, \eta) = \gamma_4 \text{rect}\left(\frac{\frac{2V^2 f_0 \eta}{c\sqrt{R_0^2 + V^2\eta^2}} - f_{\eta c}}{f_{\text{PRF}}}\right) \cdot \exp[j\theta_4(\eta)] \cdot p_4(t) \tag{13}$$

where γ_4 denotes the intensity, $p_4(t)$ denotes the fast-time term, and $\theta_4(\eta)$ denotes the slow-time varying phase.

Some interesting characteristics of RFI can be derived form (13). For example, the azimuth location of the RFI signature in the GRD image depends on the location of the RFI source. According to (13), the location of the azimuth rectangle window function, η_{ic}, can be derived by

$$Let \ \frac{2V^2 f_0 \eta_{ic}}{c\sqrt{R_0^2 + V^2 {\eta_{ic}}^2}} = f_{\eta c} \quad \rightarrow \eta_{ic}. \tag{14}$$

The central Doppler frequency $f_{\eta c}$ can be substituted by $2Vf_0\cos\theta_c/c$, where θ_c denotes the azimuth squint angle. Then, η_{ic} can be written as

$$\eta_{ic} = \frac{R_0 \tan\theta_c}{V}. \tag{15}$$

For the S-1 system where the squint angle is very small ($-0.6°$ to $0.6°$), η_{ic} is approximately zero. Note that the azimuth zero point is defined as the azimuth location of the RFI source. Therefore, (15) means that the azimuth location of the RFI pattern in the image would appear close to the actual azimuth location of the RFI source.

The azimuth extent can also be derived from (13) by

$$\begin{aligned} \eta_{iB} &= \eta_{iU} - \eta_{iL} \\ \eta_{iU} \ \text{satisfies} \ &\frac{2V^2 f_0 \eta_{iU}}{c\sqrt{R_0^2 + V^2 {\eta_{ic}}^2}} = f_{\eta c} + \frac{B_\eta}{2} \\ \eta_{iL} \ \text{satisfies} \ &\frac{2V^2 f_0 \eta_{iL}}{c\sqrt{R_0^2 + V^2 {\eta_{ic}}^2}} = f_{\eta c} - \frac{B_\eta}{2} \end{aligned} \tag{16}$$

where B_η denotes the azimuth bandwidth. Then, the azimuth extent η_{iB} can be expressed as

$$\eta_{iB} = \eta_{iU} - \eta_{iL} = \frac{R_0 \tan\theta_U}{V} - \frac{R_0 \tan\theta_L}{V} \tag{17}$$

where θ_U and θ_V denote the maximum and minimum squint angle. For the S-1 IW mode, the azimuth beam rotates from $-0.6°$ to $0.6°$, and the azimuth beamwidth is $0.23°$; therefore, the absolute values of θ_U and θ_L are no larger than $0.715°$. Since θ_U and θ_L are very small, the approximation $\tan\theta_U \approx \theta_U, \tan\theta_L \approx \theta_L$ holds, and (17) can be written as

$$\eta_{iB} \approx \frac{R_0(\theta_U - \theta_L)}{V} = \frac{R_0 \Delta\theta}{V} \approx L_{\text{syn}} \tag{18}$$

where $\Delta\theta$ is the azimuth beamwidth. Equation (18) reveals that the azimuth extent of a single RFI source η_{iB} is approximately the synthetic aperture length of the SAR system L_{syn}, whatever the modulation type of the RFI.

B. Shapes of RFI

Before we investigate the shapes of RFIs, the S-1 IW acquisition mode and the geometric relationship between SAR and RFI sources are first reviewed [see Fig. 2(a)].

The IW mode implements the Terrain Observation with Progressive Scan SAR (TOPSAR), with the antenna rotated in the azimuth direction for spotlight support. Several subswaths are acquired quasi-simultaneously by subswath switching from burst to burst [43]. If the whole GRD image is equally divided into three subimages along the range direction, then the three subimages correspond to the three subswaths (see

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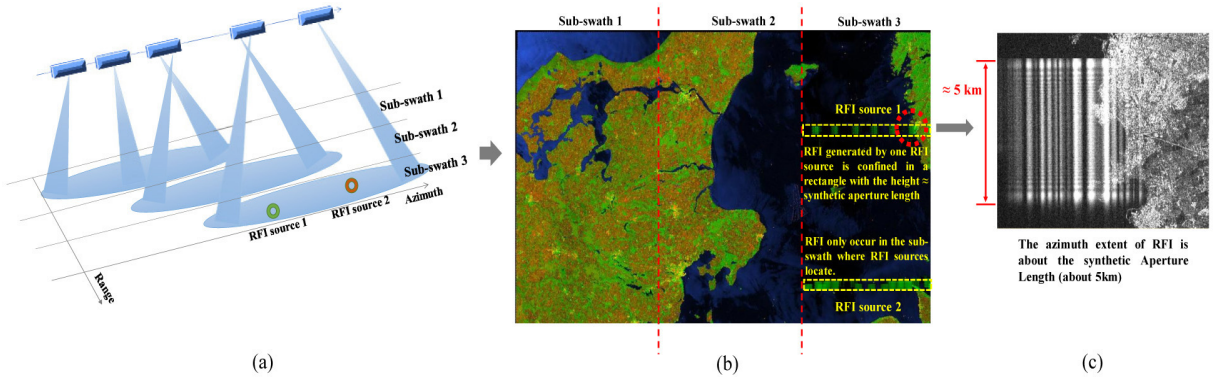


Fig. 2. IW acquisition mode of S-1. (a) Geometric relationship between SAR and RFI sources in IW mode. (b) Signatures of RFI in a GRD image. (c) Enlarged view of (b) within a synthetic aperture length.

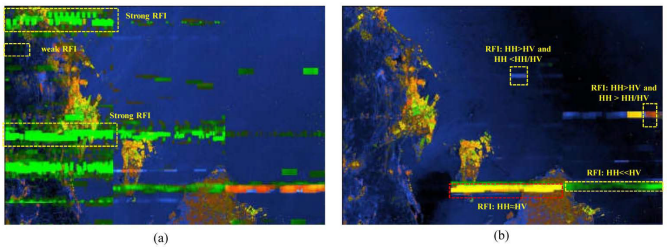


Fig. 3. Examples to show the characteristics of the intensity and polarization of RFI signatures. (a) Intensity. (b) Polarization.

Fig. 2(b) as an illustration). If there is an RFI source in one of the subswaths [e.g., Sub-swath 3 in Fig. 2(a)], the RFI signature would only occur in its corresponding subimage [see Sub-swath 3 in Fig. 2(b)].

Combining the derivation of the azimuth location in (15) and the azimuth extent in (18), in the SAR image, the azimuth location of the RFI pattern is near the actual azimuth location of the RFI source, and the azimuth extent is approximately the synthetic aperture length. Therefore, although the range extents of the RFI patterns vary with different time widths and modulation types of RFIs, the outline of the patterns of a single RFI source is always approximately a rectangle with the height of L_{syn}. As shown in Fig. 2(b), these RFIs appear as bright thin stripes in the image. As we enlarge the RFI pattern [see Fig. 2(c)], we can find that the RFI stripes are actually wide and impose a haze look on the whole synthetic aperture length (5 km). This property is very important for the RFI recognition task since it reveals that the RFI pattern is regularly shaped and such shape is quite different from the natural terrain objects in the images. Moreover, it is also an important basis for the RFI recognition methodology design in Section III.

C. Intensity of RFI

The intensities of different RFI patterns in GRD images vary widely. Some of the RFI patterns are much brighter than the terrain objects, whereas some are darker or even nearly invisible [see Fig. 3(a)]. The difference of the intensity of RFI patterns is mainly caused by the transmit power of the RFI sources and the antenna gain. If the transmit power of the RFI source is high and the transmit beam main lobe of the RFI source intersects with the receive beam main lobe of the SAR system, the RFI pattern is likely to be very bright; otherwise, it may be weak.

D. Polarization of RFI

The dual-polarization quick-look GRD images are represented by a single composite color image in RGB. The red channel (i.e., HH channel) is the co-polarization amplitude, the green channel (i.e., HV channel) is the cross-polarization amplitude, and the blue channel is the ratio between the green and the red channels [44]. As we investigate, most of the RFI patterns are bright green, which means that the intensity of RFI in the HV channel is much larger than that in the VV channel. Although not as common as the color of green, other colors of RFI patterns are also frequently observed in the image, such as red, blue, and yellow. The yellow one means that the intensity in the HH channel is comparable to that in the HV channel. Both the red and blue RFI patterns mean that the RFI is stronger in the HH channel than in the HV channel. The difference lies in that if the intensity of RFI in the HV channel is very low, then the ratio of HH to HV is larger than the absolute intensity of HH and the RFI patterns appear to be blue.

E. Characteristics of RFI in Multiple Repeat-Pass GRD Images

In the above paragraphs, we analyze the characteristics of RFI in a single GRD image. The RFI pattern shows different characteristics in terms of intensity, polarization, shapes, and textures with the terrain surface. Such difference provides possibilities for us to distinguish RFI-contaminated images from RFI-free ones. However, as there are various terrain types of the illuminated area, great differences of their signatures exist in the images of different scenarios. Cases are not rare where the signature of the terrain surface is similar to that of the RFI, which poses a great hindrance to an accurate RFI identification. An example of the repeat-pass GRD image set is given in Fig. 4, where the illuminated scene is a mountainous area on the border of Guizhou and Sichuan Province, China. We choose two images in this dataset as

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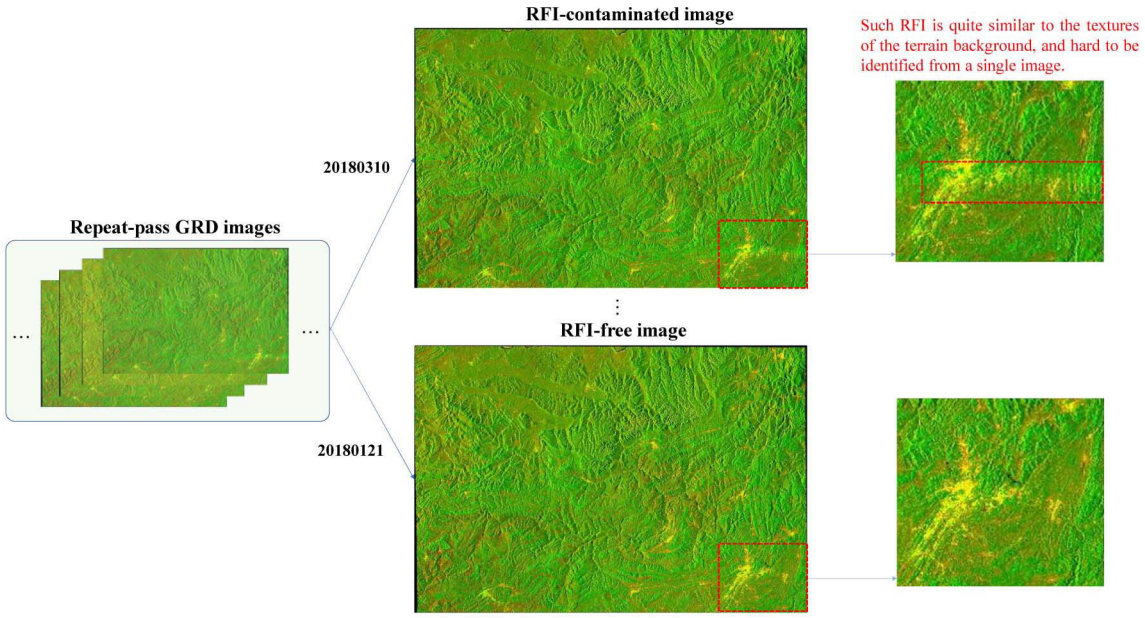


Fig. 4. Examples to show the characteristics of RFI in multiple repeat-pass GRD images (with a reference image (the bottom one), it is more easy to judge whether the top image is RFI-contaminated or not).

an illustration, one of which is RFI-free and acquired on January 21, 2018 and another of which is RFI-contaminated and acquired on March 10, 2018. As we can see from Fig. 4, the depolarization properties of the vegetation in this area result in a lot of green textures in the image. The RFI and terrain signatures are similar in the aspects of textures and shapes, colors, and intensities. From a single image, it is quite difficult to tell whether it contains RFI or not. However, if an RFI-free image is provided as a reference, the change of the textures can be observed and the RFI region could be more easily recognized.

III. Automatic RFI Identification

In this section, an automatic RFI identification method based on the Siamese-type deep CNN is proposed for S-1 signal processing. With the rapid development of deep neural networks technology [45]–[48] and its applications on remote sensing, deep learning-based methods could be introduced for RFI identification on S-1. The success of deep learning-based methods benefits from the massive high-quality supervised data, which is insufficient in the RFI identification task. Furthermore, the traditional CNN is vulnerable and not robust when the RFI is similar to the terrain backgrounds in some complex image conditions. False alarms would occur if the RFI pattern has similar features as the ground objects. To address the above problems, data augmentation is utilized to alleviate the overfitting problem and a Siamese-type deep CNN is designed to further improve the generalization ability of the RFI identification method.

A. Data Preprocessing

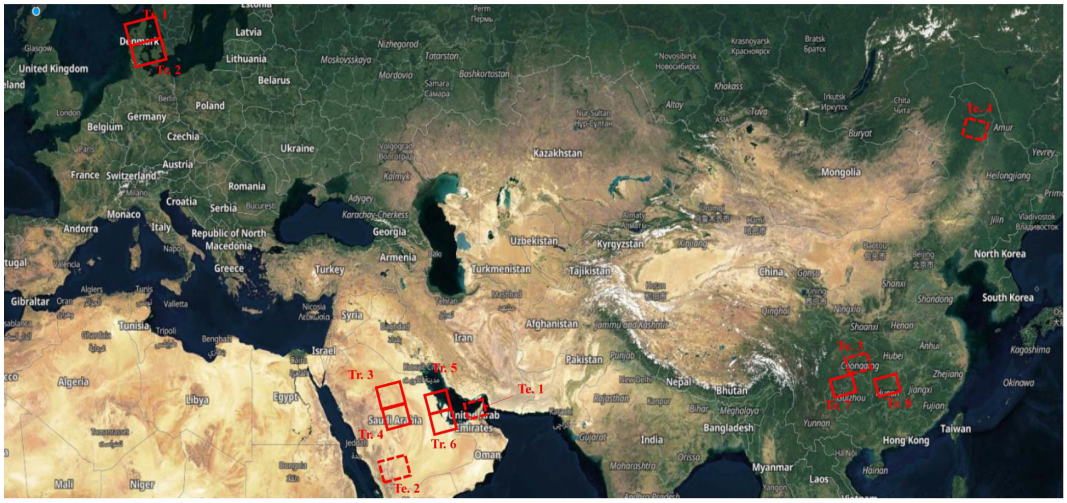
1) Scenario Selection: Dataset is prepared before introducing the automatic RFI identification model. To enable the network to capture the features of RFI, on the one hand, it is significant that the dataset should include the features of various RFIs as much as possible. The RFIs with different shapes, intensities, and characterization features have to be included in the dataset. On the other hand, the dataset should include various scenarios with different terrain types as much as possible. A comprehensive dataset that covers the possible types of RFIs and terrains is beneficial for improving the generalization ability of the method.

In this work, we establish the training dataset with eight representative scenarios around the globe with various RFI signatures and different terrain types. The test set contains four representative scenarios that are in different locations with the training set. Fig. 5 shows the locations of these scenarios in the world map. The illuminated countries and the indexes of orbits of the S-1 acquisitions are also presented in Fig. 5.

Typical RFI-free and RFI-contaminated images corresponding to the training and testing scenarios are shown in Figs. 6 and 7, respectively. The RFI sources are ground-based radar systems or other electronic equipment that operate on the same band with S-1.

As we can see from Fig. 6, the terrain background of the images in different scenarios is quite different. Training scenarios 1 and 2 lie on the border of Denmark and Sweden, most of which are bay areas with a temperate marine climate. Training scenarios 3–6 cover the area of some Middle Eastern countries, such as Saudi Arabia, Bahrain, and Qatar with a tropical dessert climate. Training scenarios 7 and 8 are mainly in the plateau area of Guizhou Province, China, with a subtropical monsoon climate. As we can see from Fig. 6, although these scenarios show different terrain backgrounds, the RFI signatures, however, show similar features (such as shapes and colors). This allows us to design a generalized method for RFI identification.

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Scenerio	Path	Frame	Location
Train 1	44	182	Border of Denmark and Sweden
Train 2	44	177	Border of Denmark and Sweden
Train 3	72	78	Riyadh, Saudi Arabia
Train 4	72	73	Riyadh, Saudi Arabia
Train 5	101	77	Border of Saudi Arabia, Bahrain and Qatar
Train 6	101	82	Border of Saudi Arabia and Qatar
Train 7	84	85	Guizhou, China
Train 8	55	87	Guizhou, China
Test 1	130	78	Dubai, United Arab Emirates
Test 2	43	53	Border of Yemen and Saudi Arabia
Test 3	157	93	Chongqing, China
Test 4	76	423	Heilongjiang, China

Fig. 5. Locations of the scenarios in the training set and test set.

As shown in Fig. 7, the test sets are also in different scenarios, where Sets 1 and 2 are bay area and Sets 3 and 4 are mountains and forests. The RFI signatures in Sets 3 and 4 are similar to the terrain backgrounds and are not as obvious as those in Sets 1 and 2. With these test sets, the generalization ability of the RFI identification network can be evaluated.

2) Data Augmentation: As shown in Fig. 6, in most cases, RFI patterns only occupy a small portion of pixels in the whole image. If the whole image is fed into the network, it is a tough task for the network to concentrate on the features of RFI under the complicated background. Therefore, the image split is performed first. The GRD image is uniformly divided into $M \times N$ subimages. After this, we augment the dataset by flipping the images horizontally and vertically. Other data augmentation methods (such as scaling and rotation) are not applied here since they would change the features of the RFI signatures. Generally, the length of RFI pattern in azimuth is comparable to the synthetic aperture length [see (18) for the derivation], while such feature would be lost after a scaling operation. Most of the RFI patterns are spread along the horizontal axis (i.e., the range axis), whereas it no longer exists after image rotation. Data augmentation step is beneficial for alleviating the overfitting problem and then enhancing the generalization performance of the network, which will be shown in the experimental analysis in Section IV.

3) Subimage Pair Formation: After data augmentation, one GRD image can yield a series of subimages. Then, these subimages are annotated with 1 (i.e., with RFI signature) or 0 (i.e., without RFI signature). Since the repeat-pass GRD image set generally contains various images corresponding to the same illuminated area. Part of these images is RFI-contaminated, whereas others are RFI-free. Therefore, for each subimage, we can form several subimage pairs by selecting RFI-free subimages in the same illuminated area as references. There are several ways to select the RFI-free reference subimage. The first one is by manual. Fortunately, for each S-1 time series, once we select the corresponding RFI-free subimage, it can be used as the reference for existing and even future subimages in the same time series. However, it is still time-consuming and manpower-wasting to search through large quantities of subimages in various imaging areas. Therefore, in this article, we use a manpower-economic method to select the RFI-free reference subimage. The conventional CNN without Siamese structure is used to predict the RFI-contaminated probability. Then, for each imaging area, the subimage with the lowest predicted RFI-contaminated probability is selected as the RFI-free reference subimage. After the reference subimage is selected for each imaging area, the subimage pairs can be formed. Each subimage pair is composed of two subimages, one of which is the RFI-free

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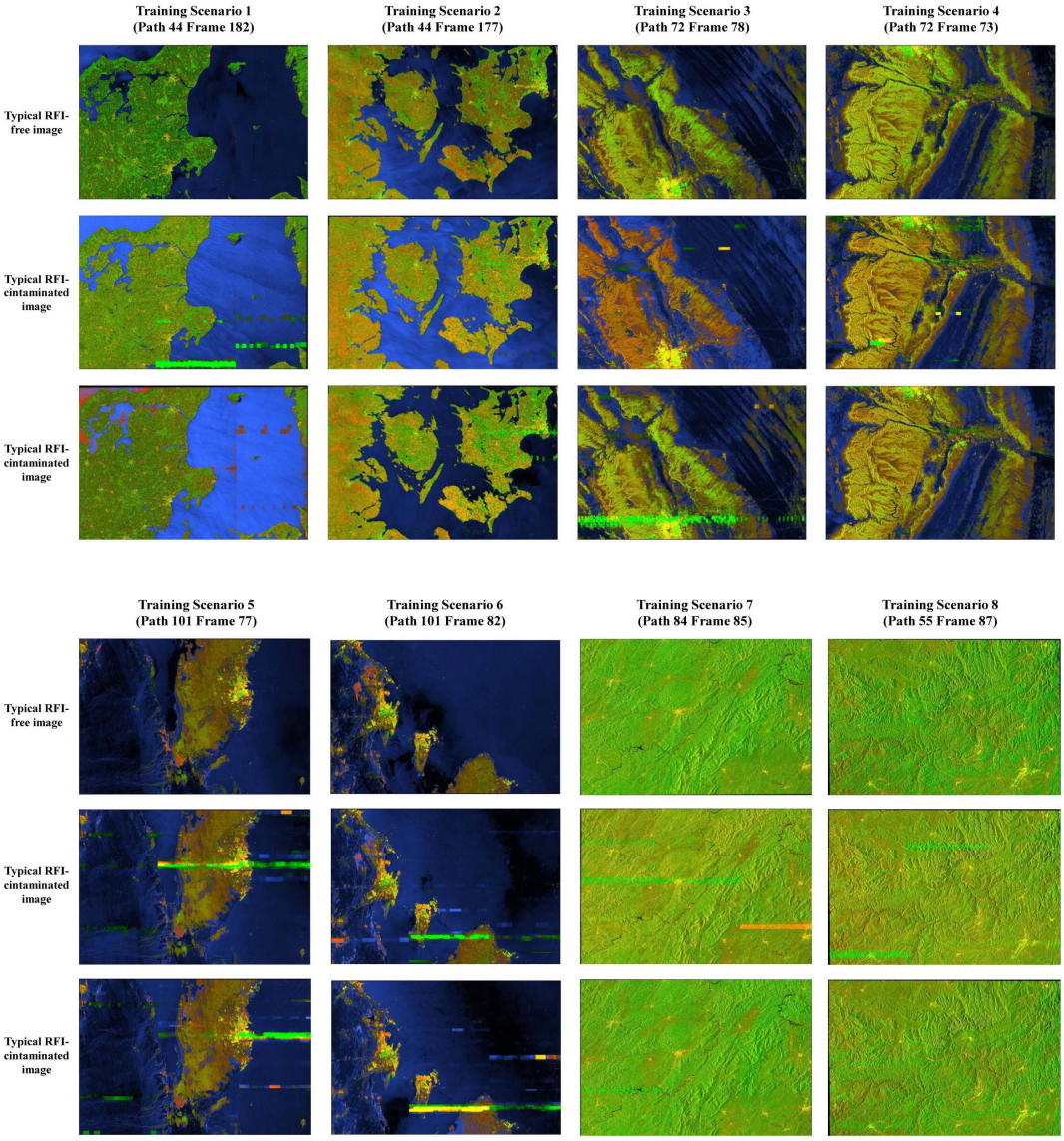


Fig. 6. Examples of the repeat-pass images corresponding to the eight scenarios in the training set.

reference subimage (i.e., Subimage A) and the other of which is the image to be verified (i.e., Subimage B). The label of the subimage pair is set as that of Subimage B. With an RFI-free image to offer a reference of the terrain background, the network can ignore the similar background features of the image pairs and pay more attention to the difference of features caused by RFI to make decisions. As such, the RFI identification performance can be improved. The process of constructing the training set, including data augmentation and subpair formation, is shown in Fig. 8.

B. RFI Identification Based on Siamese CNN

In order to resolve the background sensibility of CNN in RFI identification, a Siamese architecture [49] is imposed here. The Siamese network has been applied to various fields, such as SAR image change detection [50], face identification [49], and target tracking [51]. By comparing a pair of images, the Siamese network has the advantage of learning features related to the changes while ignoring the static background.

Considering these merits of the Siamese network, we proposed a tailor-made siamese-CNN-based network here to extract the difference between the GRD images. The architecture of the proposed Siamese CNN-based RFI identification model (Siam-CNN-RIM) is shown in Fig. 9. The input of the proposed Siam-CNN-RIM is the subimage pairs, which are generated in Section III-A3, with each pair containing two images in the same illuminated area. One of the images is a reference RFI-free image (denoted as Subimage A), and the other one is the image to be judged (denoted as Subimage B). The output is the RFI-contaminated condition of Subimage B (i.e., 1 indicates RFI-positive and 0 indicates RFI-negative). The proposed Siam-CNN-RIM is composed of two main parts: the feature extraction part to capture the feature maps of the subimage pair by a CNN and the decision-making part to discriminate whether Subimage B in the subimage pair

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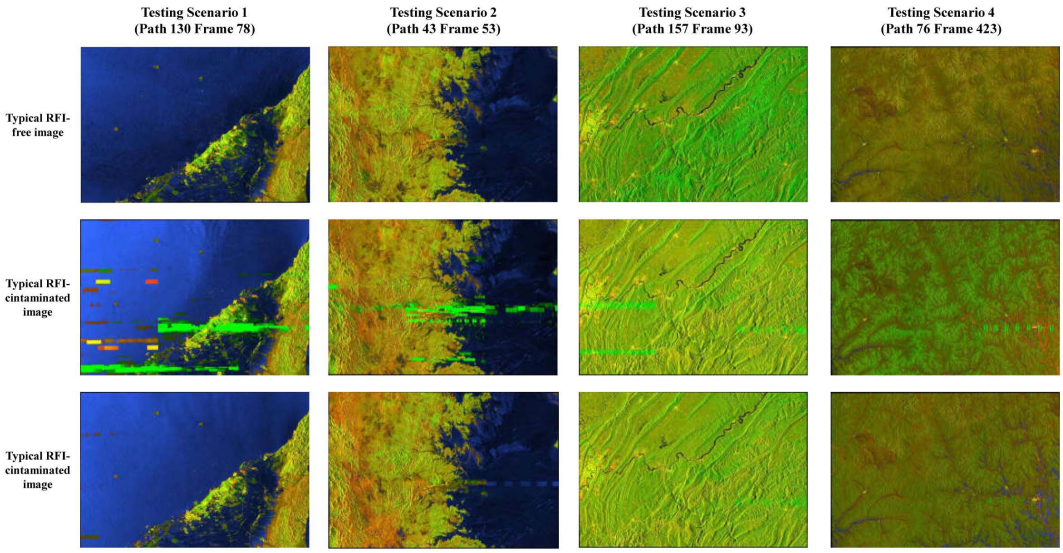


Fig. 7. Examples of the repeat-pass images corresponding to the four scenarios in the test set.

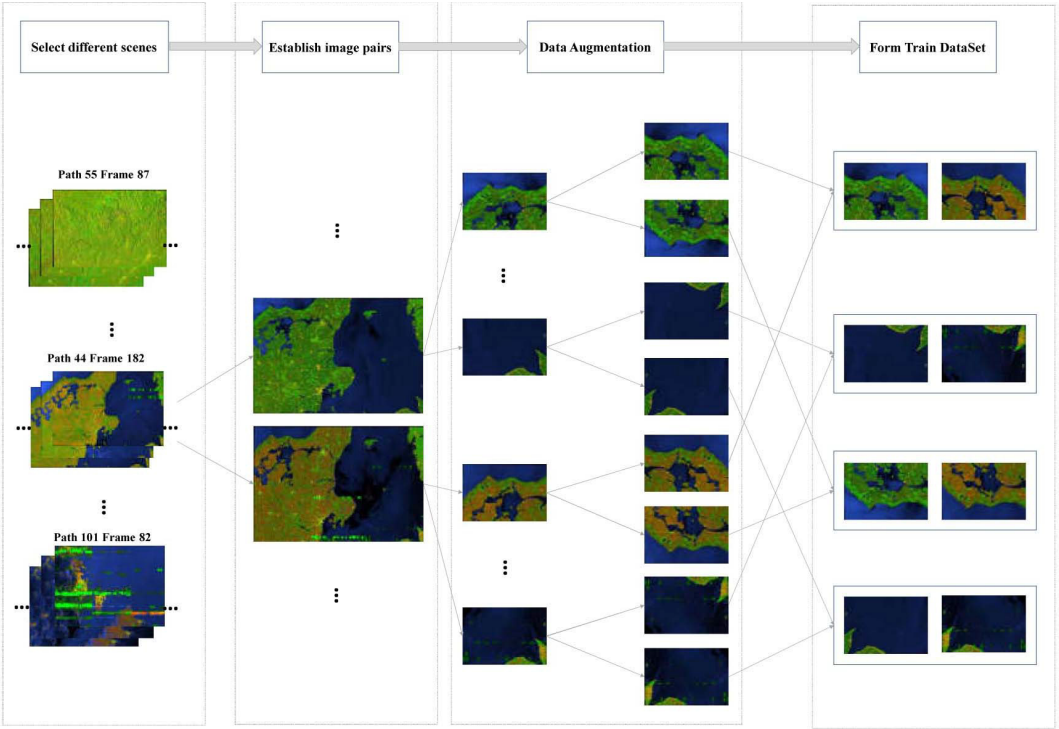


Fig. 8. Sketch diagram of constructing the training dataset.

is contaminated by RFI or not via the energy function. The diagram of the proposed Siam-CNN-RIM is shown in the top part of Fig. 9.

1) Feature Extraction: To compare the difference between different subimages more effectively, energy function is computed in the feature space rather than the high-dimensional image space. In this work, image feature is extracted via a Siamese version of multiple convolutional layers, which is composed of two identical feature extraction modules that merge into a common decision network. As such, each subimage will pass through one of these two identical CNNs that share the same weights. Each identical CNN consists of five convolutional layers that have convolution, nonlinear activation, and max-pooling operations. The ReLU function is adopted as the nonlinear activation function. Max-pooling operation works in the first, second, and fifth convolutional layers. Dropout technology is also used to alleviate the overfitting problem.

2) Decision-Making: Each subimage passed through one of the two identical feature extraction parts is flattened to a low-dimensional vector as the corresponding image feature. Then, the two image features are combined fed into the

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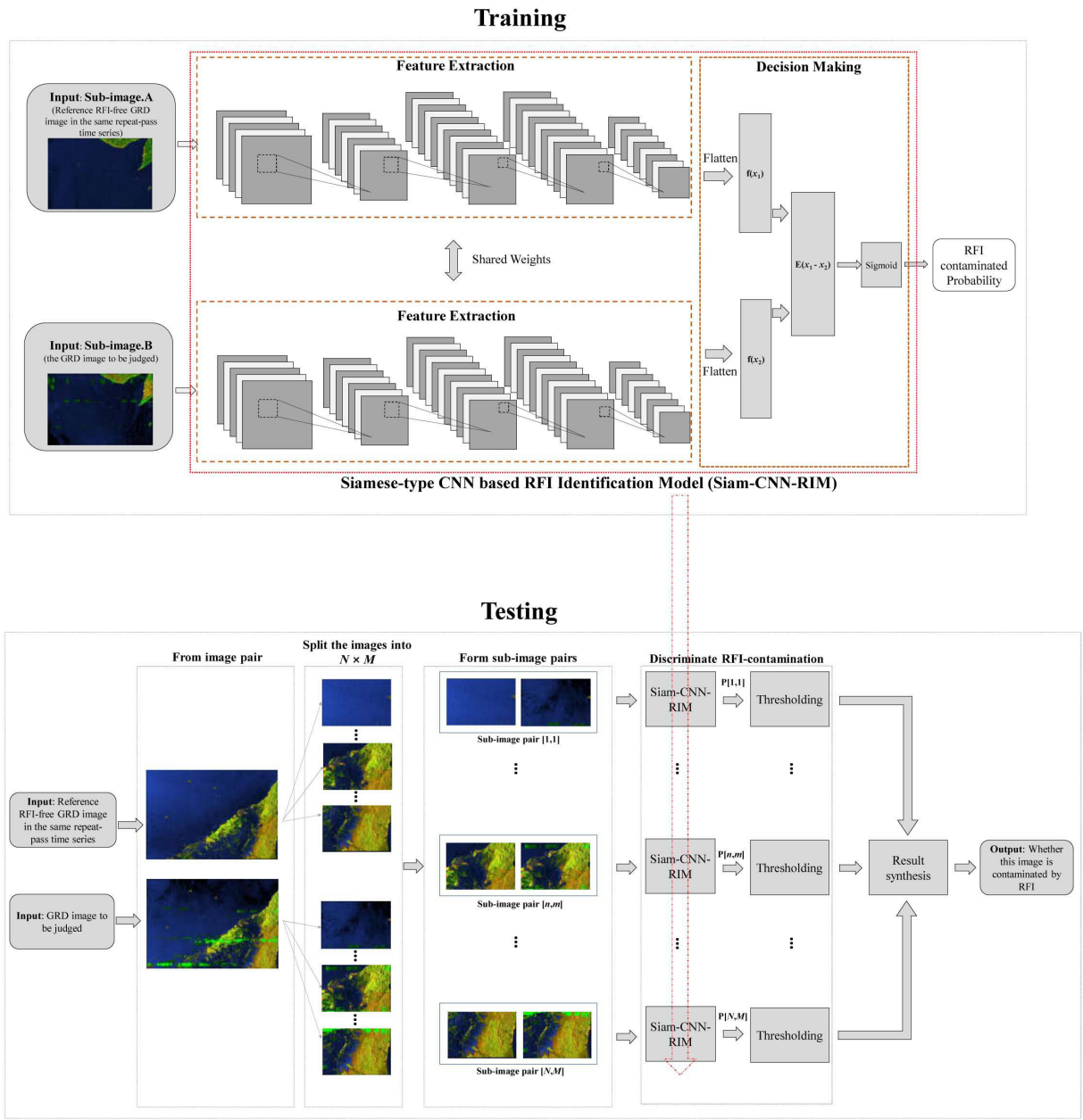


Fig. 9. Structure of the proposed Siam-CNN-RIM and the sketch diagram of the training and testing process.

energy function and get the energy value that indicates the similarity between the two subimages. The energy value is fed to the following sigmoid layer and converted to a probability between 0 and 1 that indicates whether the processed image (i.e., Subimage B in the image pair) is RFI-contaminated or not, with Subimage A as a reference of the RFI-free image.

Suppose that given a subimage pair (x_1, x_2) with label $y \in (0, 1)$, the feature extraction network will output the features $f(x_1)$ and $f(x_2)$ for each subimage. Then, we can calculate the energy function of the two features to measure the similarity between them as follows:

$$E(x_1, x_2) = \|f(x_1) - f(x_2)\|_2^2. \tag{19}$$

Here, the Euclidean distance is adopted to calculate the energy function. Therefore, the more similar the two subimages are, the smaller the energy function between them.

When training the siam-CNN, a contrastive loss function is calculated to evaluate the similarity between the two subimages x_1 and x_2 as follows:

$$L(E) = y \cdot E^2 + (1 - y)(\max(\delta - E, 0))^2 \tag{20}$$

where E is the energy function calculated in (19). y is the label of the subimage pair (x_1, x_2), $y = 1$ when the two subimages are from the same class, and $y = 0$ when the two subimages are from different classes. δ is a hyperparameter that adjusts the influence of the energy value on the contrastive loss.

C. Apply to the Open-Set Scenarios

The proposed Siam-CNN-RIM can be applied to evaluate the RFI-contaminated condition of the GRD images in other scenarios, even if for the open-set task that these scenarios

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are not included in the training set. To test a GRD image in a scenario, an RFI-free image corresponding to the same illuminated area is first selected from the repeat-pass GRD image set as a reference image. Then, an image pair is formed, which is composed of the image to be tested and the reference image. Afterward, both the images are uniformly split into $N \times M$ subimages, which are denoted as Subimage pair $[1, 1]; \ldots; [m, n]; \ldots [M, N]$, and each subimage pair is passed through the trained Siam-CNN-RIM with an output of the RFI-contaminated probability of Subimage B (i.e., $P_{1,1}; \ldots; P_{m,n}; \ldots; P_{M,N}$) in the subimage pair. A tunable thresholding step is designed to determine whether the subimage is RFI-contaminated or not. Finally, the RFI-contaminated condition of the $N \times M$ subimages of the whole GRD image is synthesized. If any subimage is judged as RFI-contaminated, then the whole image is considered to be RFI-contaminated; otherwise, the whole image is considered to be RFI-free. The diagram of the testing process is shown in the bottom part of Fig. 9.

D. Implementation Details

In practice, the architecture of the convolutional layers in the proposed Siam-CNN-RIM is similar to that of the AlexNet [45], which is composed of five convolutional layers. The first convolutional layer contains 48 filters with the size of $11 \times 11 \times 3$. The second convolutional layer contains 128 filters with the size of $5 \times 5 \times 48$. The third convolutional layer contains 192 filters with the size of $3 \times 3 \times 128$. The fourth convolutional layer contains 192 filters with the size of $3 \times 3 \times 192$. The fifth convolutional layer contains 128 filters with the size of $3 \times 3 \times 192$. In addition, max-pooling operation with the kernel size $= 3$ and stride $= 2$ is adopted in the first, second, and fifth convolutional layers.

IV. Experimental Analysis

A. Data Preparation

The S-1 GRD products are used to form the training and test datasets. The training dataset consists of eight representative scenarios around the globe with various RFI signatures and different terrain types. The test dataset consists of four other scenarios to validate the generalization performance of the proposed Siam-CNN-RIM. Details of the training and test set are available in Section III-A. The training set contains 820 images with the size of 612×900 pixels. After data augmentation ($M = 3$ and $N = 3$), these images are split into 22 140 subimages with 204×300 pixels. All these subimages are labeled with 0 (i.e., RFI-free) or 1 (i.e., RFI-contaminated). Then, for each subimage, six RFI-free subimages in the same illuminated area are selected as reference subimages to form subimage pairs. As such, 132 840 subimage pairs are obtained for the training of Siam-CNN-RIM. The test set contains 380 GRD images with 612×900 pixels, where the four test scenarios have 100, 107, 85, and 88 images.

B. Learning Curves

The training set with 132 840 subimage pairs is used to train the Siam-CNN-RIM. The model is trained on a PC with an Intel Core i5-10600KF CPU @4.10 GHz and 32-GB memory. In addition, one NVIDIA GTX 3080Ti GPU is used for acceleration. The batch size is set to be 32. The epoch number is set to be 20. Adam is selected as the optimization method, with the learning rate of 0.00002 and the weight decay of 0.0005.

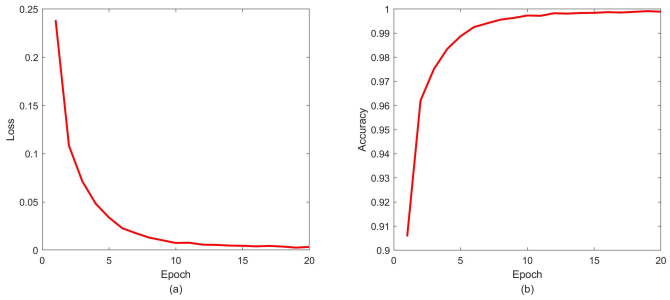


Fig. 10. Learning curves of the proposed Siam-CNN-RIM. (a) Training Loss. (b) Training accuracy.

We illustrate the evolution of our model during learning by two learning curves, one of which is calculated using the metric of loss function [see Fig. 10(a)] and the other of which is calculated using the metric of RFI identification accuracy [see Fig. 10(b)].

As we can see from Fig. 10, the proposed Siam-CNN-RIM converges very fast after about 20 epochs. The model achieves the loss close to 0 and the accuracy higher than 99% on the training set. It demonstrates that the proposed Siam-CNN-RIM well fits the training data and achieves very good performance on the training set.

C. Comparison With the State of the Art and Ablation Studies

In this section, the comparison with state-of-the-art methods and ablation studies is performed on the test set. The Lenet-type CNN-based RFI identification method proposed in [40] is used here as a benchmark. It is a pioneering work that applied deep learning to RFI identification for S-1. Compared with this work, the highlight of the proposed method includes: the deeper convolutional layer based on Alexnet, the data augmentation techniques, and the Siamese-CNN architecture. Therefore, four methods are compared for the ablation studies (see Table I). The first one is the Lenet-based CNN without the data augmentation and without the Siamese structure. The second one is the Alexnet-based CNN without the data augmentation and without the Siamese structure. The third one is the Alexnet-based CNN with the data augmentation but without the Siamese structure. The fourth method is the Alexnet-based CNN with the data augmentation and with the Siamese structure (i.e., the proposed Siam-CNN-RIM). These methods are evaluated on the test set by five metrics: the receiver operating characteristic (ROC) curve, accuracy, precision, recall, and F1-score. The ROC curve displays how well a model can classify binary outcomes. The larger the area under the ROC curve, the better the model is at distinguishing between images with RFI and without RFI. Accuracy measures how many observations are correctly classified. Precision quantifies what percentage of the positive predictions is correct. Recall

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TABLE I
ACCURACY ON THE TEST SET

Accuracy (%)	**Test set 1**	**Test set 2**	**Test set 3**	**Test set 4**	**Overall**
Lenet (w/o aug. and w/o siam.)	90.0	32.7	68.2	35.2	56.3
Alexnet (w/o aug. and w/o siam.)	81.0	95.3	71.8	71.6	80.8
Alexnet (w/ aug. and w/o siam.)	95.0	87.9	83.5	77.2	86.3
Proposed Siam-CNN-RIM	98.0	98.1	97.7	94.3	97.1

TABLE II
PRECISION ON THE TEST SET

Precision (%)	**Test set 1**	**Test set 2**	**Test set 3**	**Test set 4**	**Overall**
Lenet (w/o aug. and w/o siam.)	98.5	48.4	83.6	40.2	66.5
Alexnet (w/o aug. and w/o siam.)	82.6	98.5	79.9	81.4	85.6
Alexnet (w/ aug. and w/o siam.)	96.0	85.2	98.2	72.9	88.9
Proposed Siam-CNN-RIM	98.6	98.6	100.0	95.1	98.4

TABLE III
RECALL ON THE TEST SET

Recall (%)	**Test set 1**	**Test set 2**	**Test set 3**	**Test set 4**	**Overall**
Lenet (w/o aug. and w/o siam.)	88.0	44.3	75.0	73.8	70.2
Alexnet (w/o aug. and w/o siam.)	94.7	94.3	88.3	52.4	85.9
Alexnet (w/ aug. and w/o siam.)	85.9	70.2	97.2	90.9	91.0
Proposed Siam-CNN-RIM	98.7	98.6	97.2	92.9	97.3

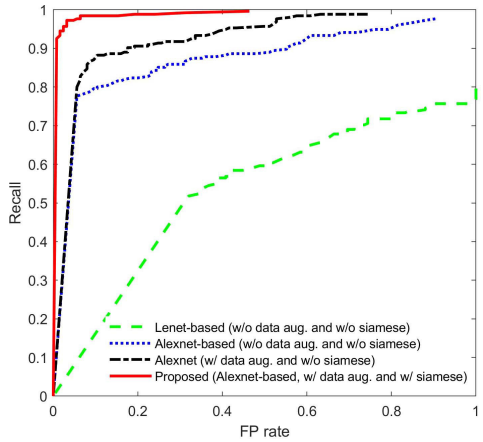


Fig. 11. ROC curve of the four methods.

quantifies what percentage of the actual positives the model is able to identify. F1-score combines precision and recall into one metric by calculating the harmonic mean between those two. The accuracy, precision, recall, and F1-score are listed in Tables II–V, respectively. The ROC curve is shown in Fig. 11.

As we can see from Fig. 11, the proposed method can achieve much higher recall than other methods under the same false positive rate. According to Tables II–V, the proposed method achieves impressive overall accuracy (97.1%), precision (98.4%), recall (97.3%), and F1-score (97.8%) in the test sets, which are significantly higher than the Lenet-based RFI identification method (accuracy: 56.3%, precision: 66.5%, recall: 70.2%, and F1-score: 68.3%). It is worth noting that, even in test sets 3 and 4 where the RFI signature is similar to the terrain background, the proposed method can still achieve satisfying performance (i.e., 97.7% and 94.3% accuracy).

The enhancement of the proposed method over existing methods is due to the imposition of a deeper convolutional layer based on Alexnet, the data augmentation techniques, and the Siamese-CNN architecture. The influences of these three factors are analyzed by ablation studies. Comparing the Alexnet (w/o aug. and w/o siam.) with the Lenet (w/o aug. and w/o siam.), the use of a deeper convolutional layer is beneficial for the feature extraction and can increase the accuracy from 56.3% to 80.8%, the precision from 66.5% to 85.6%, the recall from 70.2% to 85.9%, and the F1-score from 68.3% to 85.7%. The effectiveness of the data augmentation can be observed by comparing the Alexnet (w/ aug. and w/o siam.) with the Alexnet (w/o aug. and w/o siam.). The data augmentation step can enhance these metrics by about 5%. The performance gain of imposing a Siamese CNN structure can be seen by comparing the proposed Siam-CNN-RIM with the Alexnet (w/ aug. and w/o siam.). With a Siamese structure to extract the feature difference between two images in the same repeat-pass time series, the accuracy, precision, recall, and F1-score can be significantly increased from 86.3% to 97.1%, 88.9% to 98.4%, 91.0% to 97.3%, and 89.9% to 97.8%, respectively.

D. Influence of the Hyperparameter Selection

In this section, we investigate the influence of the hyperparameter on the performance of the proposed method. In the proposed method, a tunable threshold of probability (see Fig. 9) is used to determine whether the image contains RFI or not. The influence of the threshold on the RFI identification performance is therefore analyzed. The curves of accuracy, precision, recall, and F1-score versus different thresholds are shown in Fig. 12. As we can see in Fig. 12, the proposed method always outperforms other methods whatever the threshold value is.

There are four metrics in our experiments: accuracy, F1-score, precision, and recall. As shown in

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TABLE IV
F1-SCORE ON THE TEST SET

F1-score (%)	**Test set 1**	**Test set 2**	**Test set 3**	**Test set 4**	**Overall**
Lenet (w/o aug. and w/o siam.)	92.9	46.2	79.1	52.1	68.3
Alexnet (w/o aug. and w/o siam.)	88.2	96.4	83.3	63.8	85.7
Alexnet (w/ aug. and w/o siam.)	96.7	91.4	88.7	77.9	89.9
Proposed Siam-CNN-RIM	98.7	98.6	98.5	94.0	97.8

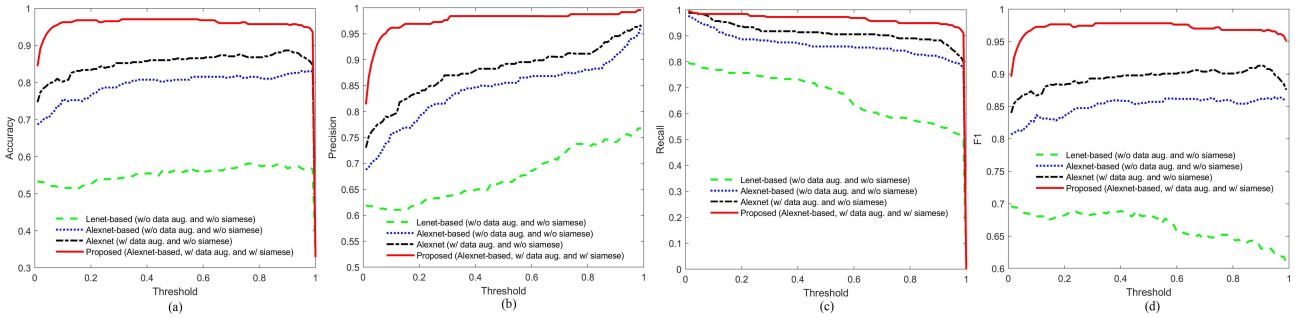


Fig. 12. Performance of the four methods for different values of threshold. (a) Accuracy. (b) Precision. (c) Recall. (d) F1-score.

Fig. 12(a) and (d), the threshold varying from 0.1 to 0.9 does not have much impact on the overall RFI judgment performance (i.e., accuracy and F1-score). However, the selection of threshold can still provide a tradeoff between precision and recall. As shown in Fig. 12(b) and (c), as the threshold increases, the precision rises and the recall decreases. Normally, a threshold of 0.5 is suggested to ensure the best overall performance and a balanced recall and precision. However, if the tradeoff between the precision and recall requires to be adjusted, the threshold can be tuned. If the users care more about the recall performance (i.e., to make sure that the RFI-contaminated images are detected as many as possible), a lower threshold (<0.5) is suggested. If the users care more about the precision performance (i.e., to ensure less false alarms that mistake the RFI-free one as RFI-contaminated), a higher threshold (>0.5) is suggested.

E. Performance When No RFI-Free References Are Available

In the proposed Siam-CNN-RIM, RFI-free subimages are required as references. Since the imaging area is revisited by S-1 for hundreds of times and the RFI sources are generally operating intermittently, cases are rare where all the revisit images are RFI-contaminated. Although such a case has not been found yet by the authors, we still suggest two ways to deal with this issue.

The first method is using CNN (i.e., the Alexnet w/ aug. and w/o siam.) directly. Method 1 can get a 86.3% RFI detection accuracy (see Table I). The second method contains three steps. The first step is to predict the RFI-contaminated probability of all the revisited images by CNN. Then, several images that are not seriously RFI-contaminated are selected and averaged to form the reference image. Finally, this reference image is used to perform Siam-CNN-RIM for more accurate RFI detection.

Comparing these two methods, with an additional Siam-CNN-RIM, Method 2 can get a slightly higher RFI detection accuracy (Method 2: 89.2% versus Method 1: 86.3%) but has a higher complexity than Method 1. Therefore, in case no RFI-free subimage is available, we suggest the users trading off the accuracy and complexity to select among these two methods.

V. DISCUSSION

Overall, our studies establish a novel Siamese-CNN-based automatic RFI identification model on the S-1 Level-1 repeat-pass GRD products. The advantage of the proposed method over existing RFI-identification methods is that it has a better generalization performance in different scenarios, even for the open-set task. The proposed method can not only identify whether a GRD image contains an RFI signature or not but also a heuristic study for other possible applications that may be beneficial for the geoscience and remote sensing society. These applications include the following.

1) RFI Mapping: Compared to previous spaceborne SAR missions, S-1 suffers from RFI not only within specific conflict zones but worldwide [52]. The mapping of RFI can show the difference of the RFI occurrence rate in different regions and different years, thus providing a reference to investigate the spatiotemporal evolution of the RFI environment. Hence, there is a strong need for RFI mapping. In the proposed Siam-CNN-RIM, the whole GRD image is first split into several subimages, and the RFI-contaminated status of each subimage is judged. Since the S-1 revisits the same place in a few days, the RFI-contaminated rate of each place can be calculated by dividing the number of RFI-contaminated subimages by the number of all the subimages corresponding to the same place. For each year and each place, the RFI-contaminated rate is calculated, and then, the spatiotemporal RFI maps can be obtained.

2) Preprocessing Before RFI Mitigation: The proposed RFI identification method can be served as a preprocessing step before RFI mitigation. Recently, various RFI mitigation methods are proposed based on the S-1 Level-1 data [11], [32]. However, how to identify the RFI-contaminated image

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has not been adequately discussed in these papers. If the RFI mitigation method is performed on the RFI-free image, the image quality would be degraded. The proposed method can be used as a preprocessing step before RFI mitigation to identify which part of the image (i.e., referred to as "subimage" in this article) contains RFI signatures. Then, we only need to perform the RFI mitigation method on the RFI-contaminated subimages and ignore those RFI-free ones. It can save a lot of computational resources and avoid the image quality degradation caused by performing the RFI mitigation method on the images that contain no RFI.

VI. Conclusion

In this article, we establish a general RFI identification model that is robust to different scenarios, even if these scenarios are not included in the training set. The proposed method can get an overall accuracy of more than 99% on the training set and get the average accuracy of 97.1%, the precision of 98.4%, the recall of 97.3%, and the F1-score of 97.8% on the test sets that are in different scenarios with the training set. These experimental results illustrate that our work provides a method for generalizing the problem of searching for the RFI artifacts. The advantage of the proposed method over traditional deep learning-based RFI identification methods lies in a deeper CNN to extract more features of RFI and the Siamese-CNN architecture to better distinguish the different signatures between RFI and ground objects. The well-designed data augmentation technique is also a plus to enhance the RFI identification performance.

This article mainly deals with RFI identification problem. For the RFI suppression problem, we have already included it as our future task and hope to enhance the RFI suppression performance by applying deep learning-based methods.

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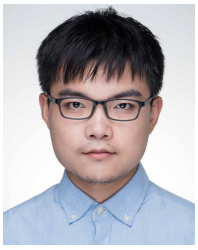
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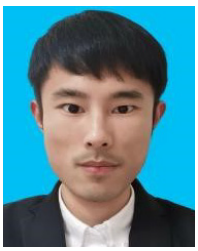
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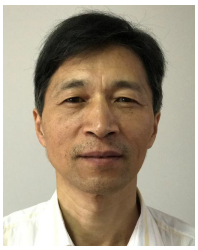
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